

September 8, 2026 Research Publication

On the Navier–Stokes Millennium Prize Problem

[Read the paper](#)

[Link to Lean formalized proof >](#)



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8:43



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We're sharing a solution to the Navier–Stokes existence and smoothness problem, one of the Millennium Prize Problems. This proof, produced by an internal OpenAI system, shows that the dynamics of the Navier-Stokes equations for fluid motion can develop a singularity in finite time. We're sharing both a writeup of the proof and a formalization in Lean.

Computational Fluid Dynamics for Aerospace Applications

Lecture 5 Discretization

Discretization

- Approximating Derivatives
- Finite Difference

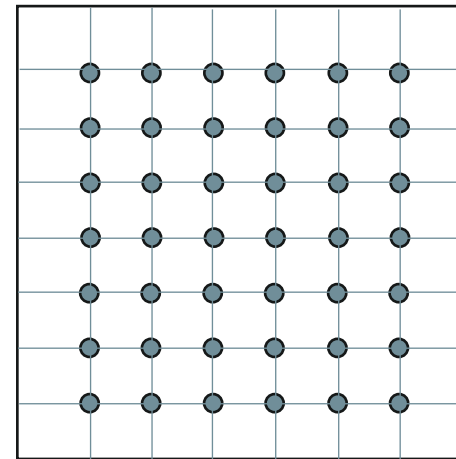
Approximating Derivatives

- We're trying to solve (or approximate) an equation like this

$$\frac{\partial(\phi)}{\partial t} + \frac{\partial(\phi u_i)}{\partial x_i} = \frac{\partial}{\partial x_i} \left(\Gamma \frac{\partial \phi}{\partial x_i} \right) + S_\phi$$

- How do we actually do it?

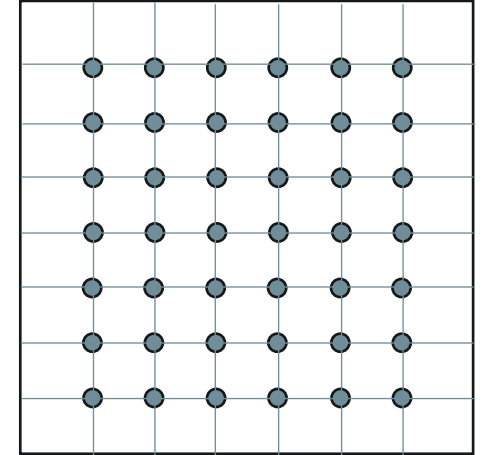
- On a grid like this



Approximating Derivatives

- There are two main approaches
- Finite Difference: Approximates derivatives that are in PDEs
 - Relatively simply to implement
 - Works well with structured grids
 - Conserving quantities takes more effort
- Finite Volume: Integrates PDEs and approximates fluxes
 - Harder to implement
 - Works well with unstructured grids
 - Conservation is fundamentally built into the approach
- We'll learn FD for simplicity, but engineering codes are often FV

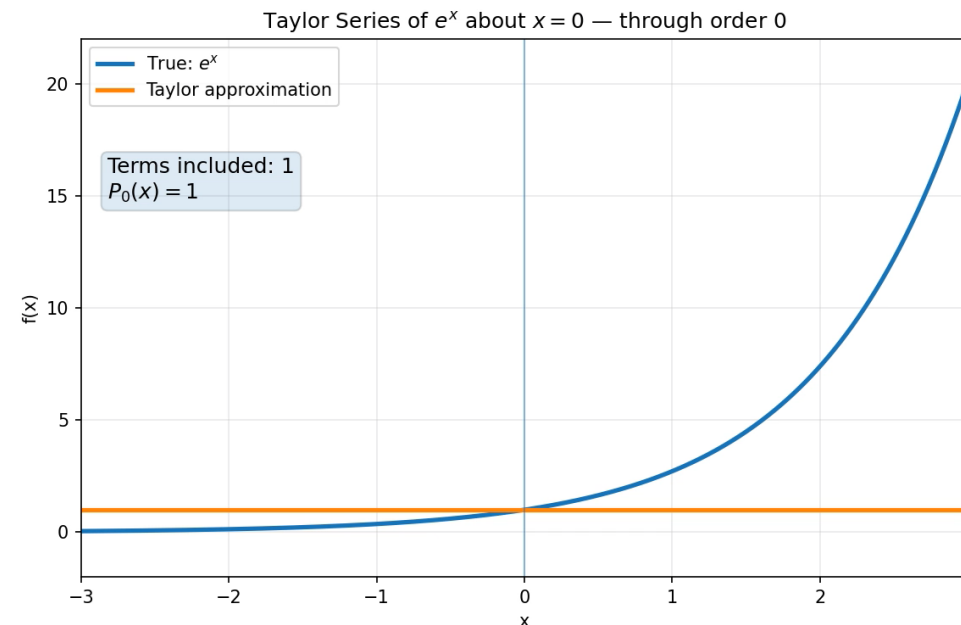
Finite Difference



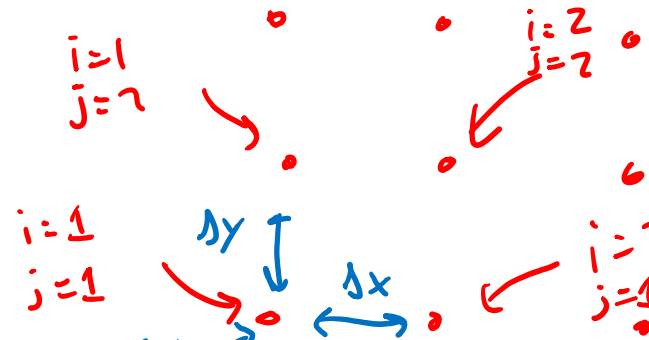
- Recall the definition of the derivative of u at $x = x_0$
- $$\frac{du}{dx} = \lim_{\Delta x \rightarrow 0} \frac{u(x_0 + \Delta x) - u(x_0)}{\Delta x}$$
- In two-dimensions, we use partial derivatives, so for point x_0, y_0 :
- $$\frac{\partial u}{\partial x} = \lim_{\Delta x \rightarrow 0} \frac{u(x_0 + \Delta x, y_0) - u(x_0, y_0)}{\Delta x}$$
- If u is continuous, we expect that this is a reasonable approximation for sufficiently small, but finite, Δx
- Essentially, we will be replacing all derivatives with such algebraic approximations and keeping Δx small

Finite Difference

- We make better approximations using Taylor Series expansion
 - Estimate the value of a function f at any point $x = x_0 + \Delta x$ as:
- $$f(x_0 + \Delta x) = f(x_0) + f'(x_0)\Delta x + \frac{f''(x_0)\Delta x^2}{2!} + \frac{f'''(x_0)\Delta x^3}{3!} + \dots$$



Finite Difference



- Let's estimate the derivative of u at $x = x_0, y = y_0$ this way:

$$f(x_0 + \Delta x) = f(x_0) + f'(x_0)\Delta x + \frac{f''(x_0)\Delta x^2}{2!} + \frac{f'''(x_0)\Delta x^3}{3!} + \dots$$

$$u(x_0 + \Delta x, y_0) = u(x_0, y_0) + \underbrace{\frac{\partial u}{\partial x} \bigg|_{x_0, y_0}}_{\text{Solve}} \Delta x + \frac{\partial^2 u}{\partial x^2} \bigg|_{x_0, y_0} \frac{\Delta x^2}{2} + \frac{\partial^3 u}{\partial x^3} \bigg|_{x_0, y_0} \frac{\Delta x^3}{6} + \dots$$

$$\underbrace{\frac{\partial u}{\partial x} \bigg|_{x_0, y_0} = \frac{u(x_0 + \Delta x, y_0) - u(x_0, y_0)}{\Delta x}}_{\text{Original definition}} - \underbrace{\left[\frac{\partial^2 u}{\partial x^2} \bigg|_{x_0, y_0} \frac{\Delta x}{2} + \frac{\partial^3 u}{\partial x^3} \bigg|_{x_0, y_0} \frac{\Delta x^2}{6} + \dots \right]}_{\text{Additional terms}}$$

$$\frac{\partial u}{\partial x} = \frac{u_{i+1,j} - u_{i,j}}{\Delta x} + \text{Error}$$

Finite Difference

$$\frac{\partial u}{\partial x} = \frac{u_{i+1,j} - u_{i,j}}{\Delta x}$$

$$+ O(\Delta x)$$

Do not
write
these

$$\left[\begin{array}{l} + O(\Delta x^2) \\ + O(\Delta x^3) \end{array} \right]$$

• Notice we have additional terms left over like $\left(\frac{\Delta x}{2} \frac{\partial^2 u}{\partial x^2} \Big|_{x_0, y_0} \right)$

• By neglecting these terms, we introduce error

- Note the error of this term is proportional to Δx
- As we go to a smaller grid, the error shrinks

• We characterize the error with “Big O” notation

• This derivative has error $O(\Delta x)$

- In the worst case, this means the error falls linearly with grid sizing
- We call this first order accurate

Finite Difference

- We can estimate this derivative in infinite ways

$$u(x - \Delta x, y) = u(x, y) + \frac{\partial u}{\partial x} (-\Delta x) + \frac{\partial^2 u}{\partial x^2} \frac{(-\Delta x)^2}{2} + \dots$$

↑
note the minus

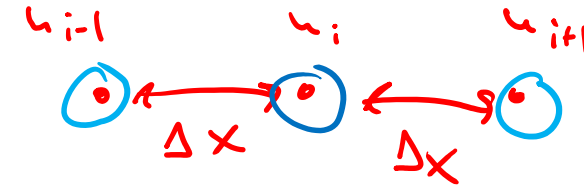
$$u(x - \Delta x, y) = u(x, y) - \underbrace{\frac{\partial u}{\partial x} \Delta x}_{\text{solve}} + \frac{\partial^2 u}{\partial x^2} \frac{\Delta x^2}{2} + \dots$$

$$\frac{\partial u}{\partial x} = \frac{u(x, y) - u(x - \Delta x, y)}{\Delta x} + \frac{\partial^2 u}{\partial x^2} \frac{\Delta x}{2} + \dots$$

$$\frac{\partial u}{\partial x} = \frac{u_{i,j} - u_{i-1,j}}{\Delta x} + O(\Delta x)$$

backward difference

Finite Difference



- We can estimate this derivative in infinite ways

Recall:

forward difference

$$\frac{\partial u}{\partial x} = \frac{u_{i+1,j} - u_{i,j}}{\Delta x}$$

$$- \frac{\partial^2 u}{\partial x^2} \frac{\Delta x}{2} - \frac{\partial^3 u}{\partial x^3} \frac{\Delta x^2}{6} + \dots$$

Backward difference

$$\frac{\partial u}{\partial x} = \frac{u_{i,j} - u_{i-1,j}}{\Delta x} + \frac{\partial^2 u}{\partial x^2} \frac{\Delta x}{2} - \frac{\partial^3 u}{\partial x^3} \frac{\Delta x^2}{6} + \dots$$

Sum both sides

$$2 \frac{\partial u}{\partial x} = \frac{u_{i+1,j} - u_{i-1,j}}{\Delta x} - \frac{\partial^3 u}{\partial x^3} \frac{\Delta x^2}{6} + \dots$$

Central difference

$$\frac{\partial u}{\partial x} = \frac{u_{i+1,j} - u_{i-1,j}}{2\Delta x} + O(\Delta x^2)$$

Finite Difference

- The three versions we've derived so far can be classified as:

- Forward Difference
$$\left. \frac{\partial u}{\partial x} \right|_{i,j} = \frac{u_{i+1,j} - u_{i,j}}{\Delta x} + O(\Delta x)$$

- Uses points ahead, first order accurate

- Backwards Difference
$$\left. \frac{\partial u}{\partial x} \right|_{i,j} = \frac{u_{i,j} - u_{i-1,j}}{\Delta x} + O(\Delta x)$$

- Uses points behind, first order accurate

- Central Difference
$$\left. \frac{\partial u}{\partial x} \right|_{i,j} = \frac{u_{i+1,j} - u_{i-1,j}}{2\Delta x} + O(\Delta x^2)$$

- Uses points around the center, second order accurate

- Each has unique mathematical properties that are important for certain applications (to be discussed)

Finite Difference

- We can also estimate higher order derivatives

Forward

$$\frac{\partial u}{\partial x} = \frac{u_{i+1,j} - u_{i,j}}{\Delta x} - \frac{\partial^2 u}{\partial x^2} \frac{\Delta x}{2} - \frac{\partial^3 u}{\partial x^3} \frac{\Delta x^2}{6} - \frac{\partial^4 u}{\partial x^4} \frac{\Delta x^3}{24} + \dots$$

Backward

$$\frac{\partial u}{\partial x} = \frac{u_{i,j} - u_{i-1,j}}{\Delta x} + \frac{\partial^2 u}{\partial x^2} \frac{\Delta x}{2} - \frac{\partial^3 u}{\partial x^3} \frac{\Delta x^2}{6} + \frac{\partial^4 u}{\partial x^4} \frac{\Delta x^3}{24} + \dots$$

Subtract (F - B)

$$0 = \frac{u_{i+1,j} - u_{i,j}}{\Delta x} - \frac{u_{i,j} - u_{i-1,j}}{\Delta x} - 2 \frac{\partial^2 u}{\partial x^2} \frac{\Delta x}{2} - 2 \frac{\partial^4 u}{\partial x^4} \frac{\Delta x^3}{24} + \dots$$

Solve

$$\frac{\partial^2 u}{\partial x^2} = \frac{u_{i+1,j} - 2u_{i,j} + u_{i-1,j}}{\Delta x^2} - \frac{\partial^4 u}{\partial x^4} \frac{\Delta x^2}{12} + \dots$$

$O(\Delta x^2)$

Finite Difference



$h = \Delta x$

- The most common forms used for first/second derivatives are given below for reference:

F $\left. \frac{\partial u}{\partial x} \right|_{i,j} = \frac{u_{i+1,j} - u_{i,j}}{h} + O(h)$

B $\left. \frac{\partial u}{\partial x} \right|_{i,j} = \frac{u_{i,j} - u_{i-1,j}}{h} + O(h)$

C $\left. \frac{\partial u}{\partial x} \right|_{i,j} = \frac{u_{i+1,j} - u_{i-1,j}}{2h} + O(h^2)$

$$\left. \frac{\partial u}{\partial x} \right|_{i,j} = \frac{-3u_{i,j} + 4u_{i+1,j} - u_{i+2,j}}{2h} + O(h^2)$$

$$\left. \frac{\partial u}{\partial x} \right|_{i,j} = \frac{3u_{i,j} - 4u_{i-1,j} + u_{i-2,j}}{2h} + O(h^2)$$

F $\left. \frac{\partial^2 u}{\partial x^2} \right|_{i,j} = \frac{u_{i,j} - 2u_{i+1,j} + u_{i+2,j}}{h^2} + O(h)$

B $\left. \frac{\partial^2 u}{\partial x^2} \right|_{i,j} = \frac{u_{i,j} - 2u_{i-1,j} + u_{i-2,j}}{h^2} + O(h)$

C $\left. \frac{\partial^2 u}{\partial x^2} \right|_{i,j} = \frac{u_{i+1,j} - 2u_{i,j} + u_{i-1,j}}{h^2} + O(h^2)$

~~$$\left. \frac{\partial^2 u}{\partial x^2} \right|_{i,j} = \frac{\delta_x^2 u_{i,j}}{h^2(1 + \delta_x^2/12)} + O(h^4)$$~~

Finite Difference

- If we're willing to use more than three points, we can go to higher derivatives (and more accurate lower derivatives)

$$h = \Delta x$$

Derivative

Finite-Difference Representation

$$\left. \frac{\partial^3 u}{\partial x^3} \right|_{i,j} = \frac{u_{i+2,j} - 2u_{i+1,j} + 2u_{i-1,j} - u_{i-2,j}}{2h^3} + O(h^2)$$

$$\left. \frac{\partial^4 u}{\partial x^4} \right|_{i,j} = \frac{u_{i+2,j} - 4u_{i+1,j} + 6u_{i,j} - 4u_{i-1,j} + u_{i-2,j}}{h^4} + O(h^2)$$

$$\left. \frac{\partial^2 u}{\partial x^2} \right|_{i,j} = \frac{-u_{i+3,j} + 4u_{i+2,j} - 5u_{i+1,j} + 2u_{i,j}}{h^2} + O(h^2)$$

$$\left. \frac{\partial^3 u}{\partial x^3} \right|_{i,j} = \frac{-3u_{i+4,j} + 14u_{i+3,j} - 24u_{i+2,j} + 18u_{i+1,j} - 5u_{i,j}}{2h^3} + O(h^2)$$

$$\left. \frac{\partial^2 u}{\partial x^2} \right|_{i,j} = \frac{2u_{i,j} - 5u_{i-1,j} + 4u_{i-2,j} - u_{i-3,j}}{h^2} + O(h^2)$$

$$\left. \frac{\partial^3 u}{\partial x^3} \right|_{i,j} = \frac{5u_{i,j} - 18u_{i-1,j} + 24u_{i-2,j} - 14u_{i-3,j} + 3u_{i-4,j}}{2h^3} + O(h^2)$$

$$\left. \frac{\partial u}{\partial x} \right|_{i,j} = \frac{-u_{i+2,j} + 8u_{i+1,j} - 8u_{i,j} + u_{i-1,j}}{12h} + O(h^4)$$

$$\left. \frac{\partial^2 u}{\partial x^2} \right|_{i,j} = \frac{-u_{i+2,j} + 16u_{i+1,j} - 30u_{i,j} + 16u_{i-1,j} - u_{i-2,j}}{12h^2} + O(h^4)$$

Finite Difference

- And we can even perform such analysis for mixed partial derivatives

$$\frac{\partial}{\partial x} \left(\frac{\partial u}{\partial y} \right)$$

$$= \frac{\partial}{\partial x} \left(\frac{u_{j+1} - u_{j-1}}{2\Delta y} \right) \quad \frac{\partial A}{\partial x} = \frac{A_{i+1} - A_{i-1}}{2\Delta x}$$

A

Derivative

Finite-Difference Representation

$$\left. \frac{\partial^2 u}{\partial x \partial y} \right|_{i,j} = \frac{1}{\Delta x} \left(\frac{u_{i+1,j} - u_{i+1,j-1}}{\Delta y} - \frac{u_{i,j} - u_{i,j-1}}{\Delta y} \right) + O(\Delta x, \Delta y)$$

$$\left. \frac{\partial^2 u}{\partial x \partial y} \right|_{i,j} = \frac{1}{\Delta x} \left(\frac{u_{i,j+1} - u_{i,j}}{\Delta y} - \frac{u_{i-1,j+1} - u_{i-1,j}}{\Delta y} \right) + O(\Delta x, \Delta y)$$

$$\left. \frac{\partial^2 u}{\partial x \partial y} \right|_{i,j} = \frac{1}{\Delta x} \left(\frac{u_{i,j} - u_{i,j-1}}{\Delta y} - \frac{u_{i-1,j} - u_{i-1,j-1}}{\Delta y} \right) + O(\Delta x, \Delta y)$$

$$\left. \frac{\partial^2 u}{\partial x \partial y} \right|_{i,j} = \frac{1}{\Delta x} \left(\frac{u_{i+1,j+1} - u_{i+1,j}}{\Delta y} - \frac{u_{i,j+1} - u_{i,j}}{\Delta y} \right) + O(\Delta x, \Delta y)$$

$$\left. \frac{\partial^2 u}{\partial x \partial y} \right|_{i,j} = \frac{1}{\Delta x} \left(\frac{u_{i+1,j+1} - u_{i+1,j-1}}{2\Delta y} - \frac{u_{i,j+1} - u_{i,j-1}}{2\Delta y} \right) + O[\Delta x, (\Delta y)^2]$$

$$\left. \frac{\partial^2 u}{\partial x \partial y} \right|_{i,j} = \frac{1}{\Delta x} \left(\frac{u_{i,j+1} - u_{i,j-1}}{2\Delta y} - \frac{u_{i-1,j+1} - u_{i-1,j-1}}{2\Delta y} \right) + O[\Delta x, (\Delta y)^2]$$

$$\left. \frac{\partial^2 u}{\partial x \partial y} \right|_{i,j} = \frac{1}{2\Delta x} \left(\frac{u_{i+1,j+1} - u_{i+1,j-1}}{2\Delta y} - \frac{u_{i-1,j+1} - u_{i-1,j-1}}{2\Delta y} \right) + O[(\Delta x)^2, (\Delta y)^2]$$

$$\left. \frac{\partial^2 u}{\partial x \partial y} \right|_{i,j} = \frac{1}{2\Delta x} \left(\frac{u_{i+1,j+1} - u_{i+1,j}}{\Delta y} - \frac{u_{i-1,j+1} - u_{i-1,j}}{\Delta y} \right) + O[(\Delta x)^2, \Delta y]$$

$$\left. \frac{\partial^2 u}{\partial x \partial y} \right|_{i,j} = \frac{1}{2\Delta x} \left(\frac{u_{i+1,j} - u_{i+1,j-1}}{\Delta y} - \frac{u_{i-1,j} - u_{i-1,j-1}}{\Delta y} \right) + O[(\Delta x)^2, \Delta y]$$

error can differ

*

Finite Difference

- Approximations with higher-order accuracy require evaluation of more points and so run slower/require more compute time
- However, since they are more accurate, you may not need as many grid points to obtain a good answer
- Higher-order accuracy approximations are better at handling discontinuities (e.g., shockwaves)
- Note that most real CFD codes aim for second-order or higher

Finite Difference

- What about derivatives in time?
- We can largely do the same thing with Taylor Series
 - Note that “backward differencing” doesn’t really make sense for time
- For example, Explicit Euler methods are basically just first order forward difference but in time

n is time index

$$\left. \frac{\partial u}{\partial t} \right|_{i,j,n} = \frac{u_{i,j}^{n+1} - u_{i,j}^n}{\Delta t} + O(\Delta t)$$

Finite Difference

- Bringing it all together, let's approximate a simple PDE
- One-dimensional unsteady advection equation

$$\frac{\partial u}{\partial t} + c \frac{\partial u}{\partial x} = 0$$

- Why solve this equation?
- Note its similarity to the LHS of x-momentum in Euler Equations

$$\frac{\partial(\rho u)}{\partial t} + \frac{\partial(\rho u u)}{\partial x} + \cancel{\frac{\partial(\rho u v)}{\partial y}} + \cancel{\frac{\partial(\rho u w)}{\partial z}} = -\frac{\partial p}{\partial x} + \cancel{\rho f_x}$$

Finite Difference

Explicit Euler (time)
First Order Forward Difference (space)

$$\frac{\partial u}{\partial t} + c \frac{\partial u}{\partial x} = 0$$

Exclude moving forward (but still here)

$$\frac{u_i^{n+1} - \hat{u}_i^n}{\Delta t} + c \left[\frac{\hat{u}_{i+1}^n - \hat{u}_i^n}{\Delta x} \right] + O(\Delta t) + O(\Delta x) = 0$$

$$u_i^{n+1} = \underbrace{\hat{u}_i^n}_{\text{current time } n} - \frac{c \Delta t}{\Delta x} \left[\frac{\hat{u}_{i+1}^n - \hat{u}_i^n}{\Delta x} \right]$$

↑
next time n+1

Finite Difference

- $u_j^{n+1} = u_j^n - \frac{c\Delta t}{\Delta x} (u_{j+1}^n - u_j^n)$
- Solve this equation at each point at time $n + 1$ to update the solution in time



initial conditions

Looking Forward

- We derived *an* approximation to the equation
- What makes a good approximation?
 - Order of accuracy is part of the story
 - We are first order in time and space, about as bad as it can be but “correct”
 - Stability
 - Solving the correct equations with any arbitrary approximation does not guarantee the answer will trend towards the correct answer
 - Boundary Conditions
 - What is happening at the edges of our box? Do we describe it well?